

ECE 447

Lesson 31

Probability, Part 2



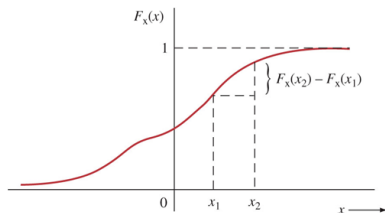
UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE AND ADMIN

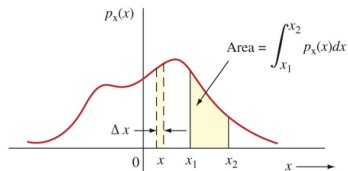
- [Schedule](#)
- Admin
 - **Lab 6.** PDF due tonight (Lesson 31, 4 Nov) to Gradescope.
 - **HW7.** Due Lesson 32 (Fri, 6 Nov) to Gradescope.
 - **HW8.** Assigned today. Due Lesson 34 (Fri, 13 Nov) to Gradescope.
 - **GR2** is Lesson 35 (Tue, 17 Nov), covering Lessons 18-32 (Chapters 5-8). Review day is Lesson 34 (Fri, 13 Nov).

CDF AND PDF

- CDF: $F_X(x) = P(X \leq x)$
 - $F_X(x) \geq 0$
 - $F_X(\infty) = 1$
 - $F_X(-\infty) = 0$
 - $F_X(x_1) \leq F_X(x_2)$ for $x_1 \leq x_2$
- PDF: $p_X(x) = \frac{dF_X(x)}{dx}$
 - $\int_{-\infty}^{\infty} p_X(x) dx = 1$
 - $P(x_1 < x \leq x_2) = \int_{x_1}^{x_2} p_X(x) dx$



(a)

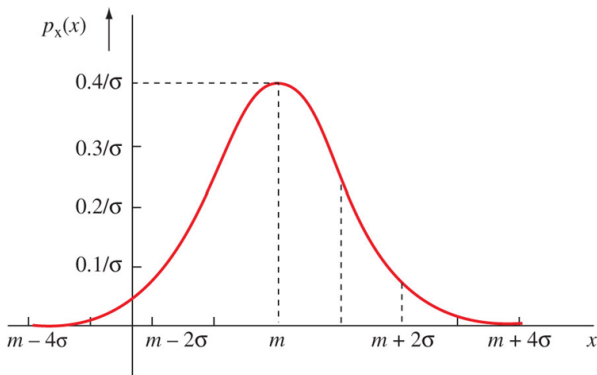


(b)

GAUSSIAN RV

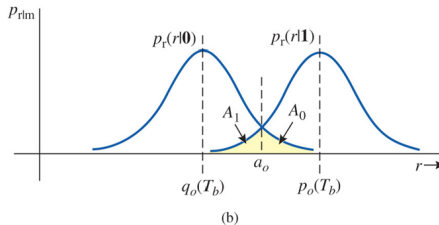
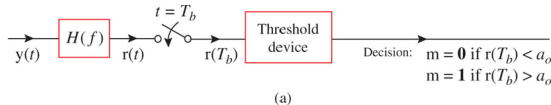
- Standard Gaussian has zero mean and unit variance, $x \sim \mathcal{N}(0, 1)$
- General equation for $x \sim \mathcal{N}(m, \sigma)$ is $p_x(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-(x-m)^2/(2\sigma^2)}$
- Typically evaluate using $Q(\cdot)$ functions - area under curve tail

$$P(x > x) = Q\left(\frac{x - m}{\sigma}\right)$$



GAUSSIAN RV IN COMMUNICATION SYSTEMS

- Recall (Lsn 25): only care about sample value (test statistic) at the receiver
- $r(T_b) = s_k(T_b) + n(T_b) \rightarrow s_k(T_b)$ deterministic, $n(T_b) \sim \mathcal{N}(0, \sigma)$
- How is $r(T_b)$ distributed? We need to know how to find the *expected value*, $\mathbb{E}[r(T_b)]$
- We'll come back to this problem...



EXPECTED VALUE

- Aka average value aka mean value
- **For discrete RV:** $\bar{x} = \mathbb{E}[x] = \sum_i x_i P_x(x_i)$
- **For continuous RV:** $\bar{x} = \mathbb{E}[x] = \int_{-\infty}^{\infty} x p_x(x) dx$
- **Mean of a sum:** mean of the sum is equal to the sum of the means, $\overline{x + y} = \bar{x} + \bar{y}$
- **Mean of a product:** messy, unless independent - then $\overline{xy} = \bar{x} \cdot \bar{y}$

MOMENTS

- The n th moment defined as the mean value of x^n
- **First moment:** expected value, $\mathbb{E}[x] = \int_{-\infty}^{\infty} xp_x(x)dx$
- **Central moments:** subtract out the mean first,
 $\mathbb{E}^n[(x - \bar{x})^n] = \int_{-\infty}^{\infty} (x - \bar{x})^n p_x(x)dx$
- **Second central moment:** variance, mean square value minus square of the mean:

$$\sigma_x^2 = \mathbb{E}^2[(x - \bar{x})^2] = \int_{-\infty}^{\infty} (x - \bar{x})^2 p_x(x)dx = \overline{x^2} - \bar{x}^2$$

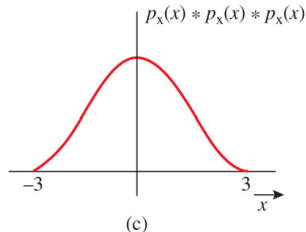
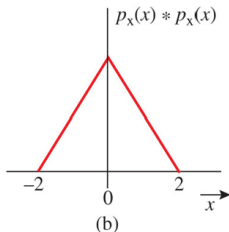
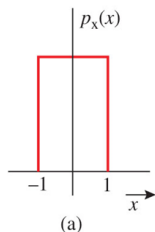
- Variance of a sum of independent RVs is equal to the sum of their variances
 - If x and y are independent RVs and $z = x + y$, then $\sigma_z^2 = \sigma_x^2 + \sigma_y^2$

PRACTICE: Q FUNCTION AND MOMENTS

- (a) Reading a Gaussian.** Detector noise $n \sim \mathcal{N}(0, \sigma)$ with $\sigma = 0.5$ V, threshold at 0.9 V.
 - $P(n > 0.9) = Q\left(\frac{x-m}{\sigma}\right) = Q(\text{____}) = \text{____}$
 - Now the noise picks up a 0.2 V DC offset. Up or down, and by how much?
 - $P(|n| > 0.9) = \text{____}$ — why is that not the same as $P(n > 0.9)$?
 - Handed $p_x(x) = \frac{1}{C\sqrt{2\pi}}e^{-(x-1)^2/8}$, read off $m = \text{____}$ $\sigma = \text{____}$ $C = \text{____}$
- (b) Moments.** Quantization error is uniform on $\left[-\frac{\Delta}{2}, \frac{\Delta}{2}\right]$, so $p_x(x) = 1/\Delta$ there.
 - $\mathbb{E}[x] = \text{____}$ by inspection — why? $\mathbb{E}[x^2] = \int_{-\Delta/2}^{\Delta/2} \frac{x^2}{\Delta} dx = \text{____}$
 - $\sigma_x^2 = \overline{x^2} - \bar{x}^2 = \text{____}$ — the PCM quantization noise power from Lesson 19
- HW8 leans on both halves; the mean-square step returns in Lesson 36 as noise power.

CENTRAL LIMIT THEOREM

- The sum of a large number n independent RVs with mean μ and variance σ^2 tends to be a Gaussian RV, regardless of the PDFs of the variables added
- Gaussian mean is $n\mu$
- Gaussian variance is σ^2/n
- Example 7.25 (similar to future HW problem, 7.7-1)
 - Note the problem defines a discrete RV, x_i
 - Use $\bar{x} = \mathbb{E}[x] = \sum_i x_i P_x(x_i)$



GAUSSIAN RV IN COMMUNICATION SYSTEMS

- Back to this problem!
- $r(T_b) = s_k(T_b) + n(T_b) \rightarrow s_k(T_b)$ deterministic, $n(T_b) \sim \mathcal{N}(0, \sigma)$
- $\mathbb{E}[r(T_b)] = \mathbb{E}[s_k(T_b) + n(T_b)] =$
- $\sigma_r^2 = \mathbb{E}[r^2(T_b)] - \mathbb{E}[r(T_b)]^2 =$

